

Pre-Board Examination - 2079

Class : XII

Subject: Biology (Code: 2021)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt All Questions

PART : I [BOTANY]

Group 'A' [BOTANY]

Rewrite the best alternative of each question in your answer sheet.

[5 × 1 = 5]

1. Apical dominance is caused by
 - a. Auxin
 - b. Gibberellins
 - c. Ethylene
 - d. Cytokinins
2. The green colour-blindness is called
 - a. Protanopia
 - b. Deuteranopia
 - c. Tritanopia
 - d. Hypermetropia
3. Ginger plant is propagated by
 - a. Seeds
 - b. Tubers
 - c. Rhizomes
 - d. Fruits
4. The another name for epiblema is
 - a. Skin
 - b. Epidermis
 - c. Exodermis
 - d. Piliferous layer
5. Many RNA primers are formed in
 - a. Lagging strand
 - b. Leading strand
 - c. Template strand
 - d. Okazaki fragment

Group 'B' [BOTANY]

Give short answer to the following questions.

[4 × 4 = 16]

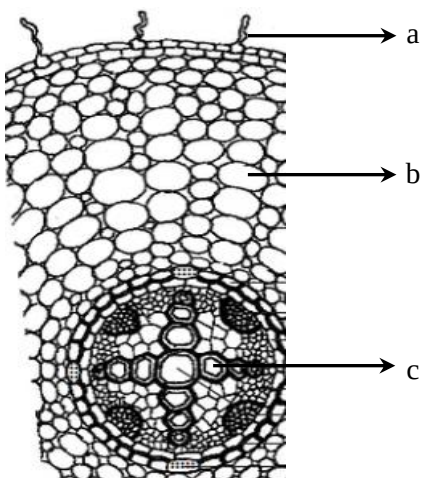
1. What is green manure? Write down its significance. [1+3 = 4]
2. Define entomophily. Mention the characters of entomophilous plants. [1+3 = 4]
3. Write down the anatomical differences between dorsiventral and isobilateral leaves. [4]

OR

What is vascular bundle? Write down its types with necessary sketches

4. Answer the questions based on the given diagram

- a. Give the name of the figure
- b. Name the parts a, b and c



Group 'C' [BOTANY]

Give long answer to the following questions.

[2 × 8 = 16]

5. Explain Watson and Crick model of DNA. Write down the functions of DNA.

[6+2 = 8]

OR

What are sex-linked genes? How many types of sex-linked genes are there?

Show the inheritance pattern of eye colour in *Drosophila*.

[1+1+6 = 8]

6. What is the first stable product of photosynthesis? Describe the light independent steps of photosynthesis.

[1+7 = 8]

PART : II [ZOOLOGY]

Group 'A' [ZOOLOGY]

Rewrite the best alternative of each question in your answer sheet.

[6 × 1 = 6]

1. Camels hump is made up of
 - a. Skeletal tissue
 - b. Muscular tissue
 - c. Adipose tissue
 - d. Bone
2. Gray crescent in frog's egg marks the future:
 - a. Dorsal side of the embryo
 - b. Ventral side of the embryo
 - c. Lateral side of the embryo
 - d. Tail of the embryo
3. How many teeth in man grows twice in life?
 - a. 32
 - b. 28
 - c. 20
 - d. 12
4. Synthesis of urea in liver takes place by
 - a. Nitrogen cycle
 - b. Krebs cycle
 - c. Ornithine cycle
 - d. Glycolysis
5. Parathormone induces
 - a. Increase in serum calcium level
 - b. Decrease in serum calcium level
 - c. Increase in blood sugar level
 - d. Decrease in blood sugar level
6. Passive immunity is defined as immunity
 - a. Inherited from the parents
 - b. Achieved through vaccination
 - c. Acquired through first exposure to the
 - d. Achieved through the sera of other animals' disease enriched in antibodies

Group 'B' [ZOOLOGY]

Give short answers to the following questions.

[4 × 4 = 16]

1. Describe the process of fertilization of an egg with reference to frog. [4]

OR

What is vaccine? Describe the various types of vaccine. [1+3 = 4]

2. How are the following enzymes activated in the alimentary canal? [1×4 = 4]

i. Trypsinogen

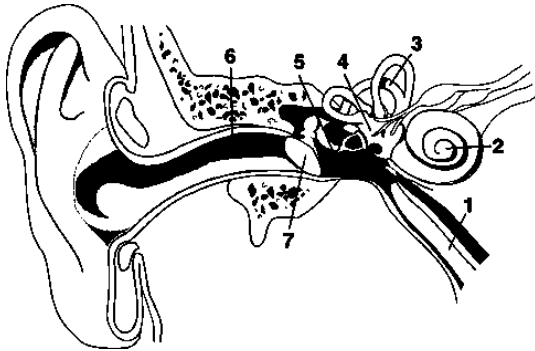
ii. Chymotrypsinogen

iii. Pepsinogen

iv. Prorennin

3. Mention any four measure differences between voluntary and involuntary muscle fibres. [1×4 = 4]

4. Give the names of labelled numbers 2, 3, 6 and 7 in the given figure. [4]



Group 'C' [ZOOLOGY]

Give long answers to the following questions.

[2 × 8 = 16]

5. Describe the internal structure of human heart with well labelled diagram. [5+3 = 8]

OR

What is digestion? Draw a well-labelled diagram of human alimentary canal and describe its different parts. [1+2+5 = 8]

6. Are HIV and AIDS the Same Thing? Which Activities Are Most Likely to Transmit HIV? What is the main problem of AIDS? Mention the diagnostic test.

[2+3+2+1 = 8]

The End